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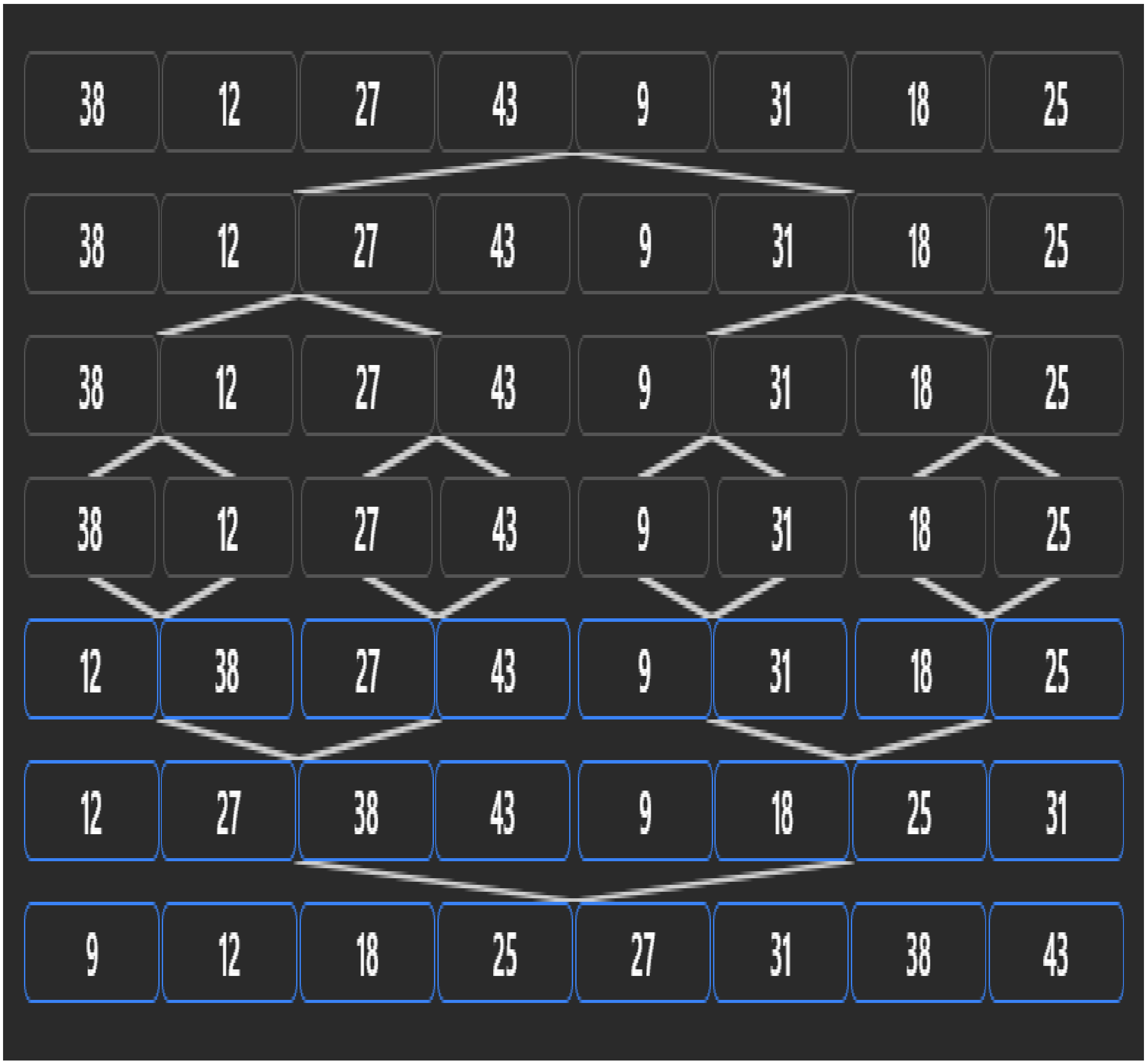
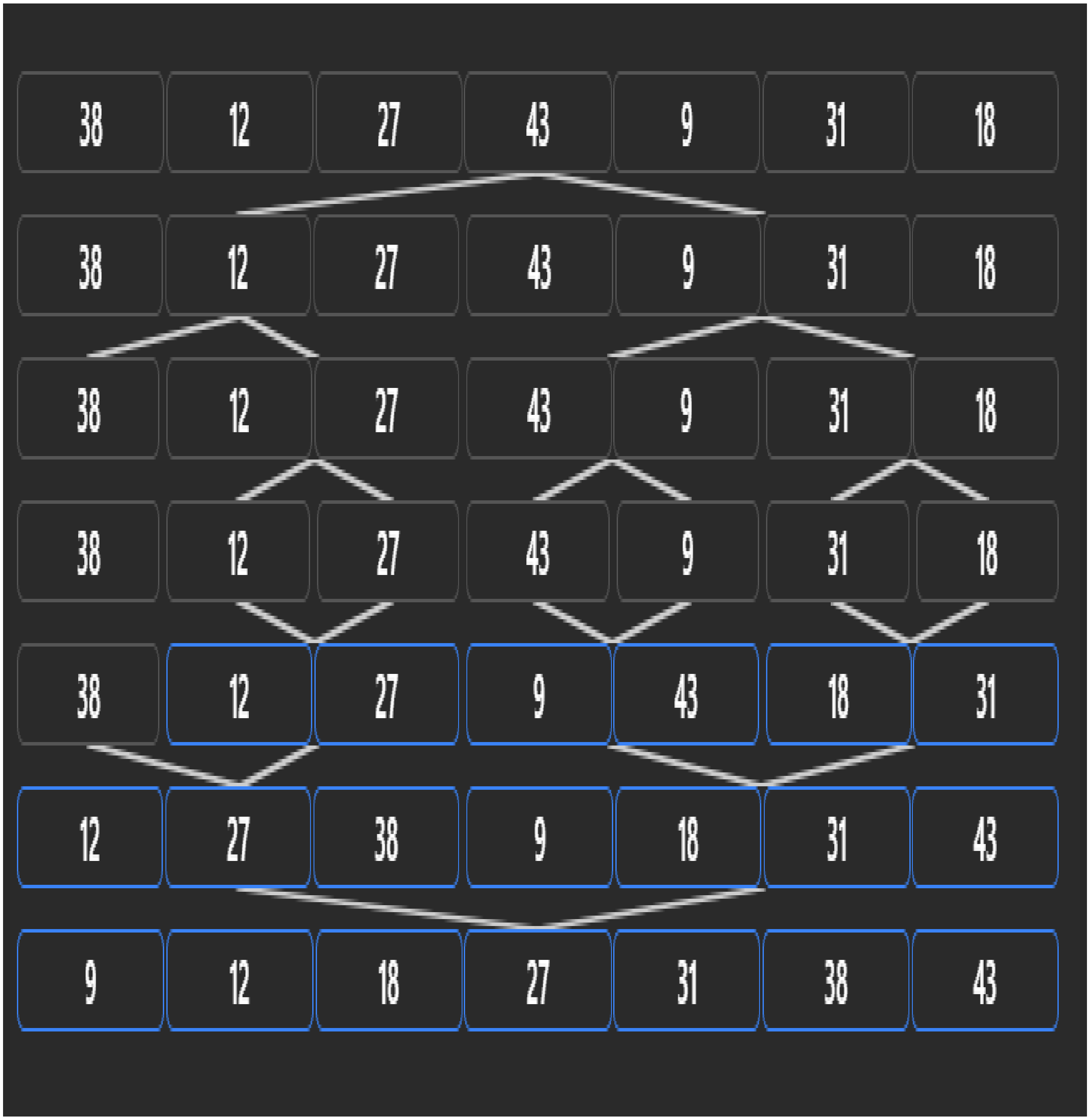
Below is a **simple complete C program using OpenMP** that satisfies all 5 requirements. It also includes a **hashing strategy** for efficient data storage and **parallel merge sort** for arranging the data.

8-EVEN



7-ODD



38 12 27 43 9 31 18
38 12 27 43 9 31 18
38 12 27 43 9 31 18
38 12 27 43 9 31 18
38 12 27 9 43 18 31
12 27 38 9 18 31 43
9 12 18 27 31 38 43

Algorithm

- Generate/store integer data in an array.
- Use a hash function key % TABLE_SIZE for efficient storage/indexing.
- Apply recursive merge sort.
- If the sub array size is less than **10000**, use insertion sort.
- Create Open MP tasks for the left and right halves.
- Use #pragma omp task wait before merging.
- Start parallel execution using #pragma omp parallel and #pragma omp single.
- Measure execution time for **1, 2, 4, and 8 threads**.
- Calculate speedup using:

Complexity

Operation	**Complexity**
Hashing	Average **$O(1)$**
Insertion Sort	**$O(n^2)$** worst case
Merge Sort	**$O(n \log n)$**
Parallel Merge Sort	Approximately **$O((n \log n)/p)$**
Extra Space	**$O(n)$**